1 I

1.1 (1)

The kinetic energy is:

$$T = \frac{1}{2} m l^2 \dot{\theta}^2,$$

and the potential energy is:

$$V = -mgl\cos\theta \approx -mgl\left(1 - \frac{\theta^2}{2}\right).$$

Thus, the Lagrangian is:

$$L = T - V \approx \frac{1}{2} m l^2 \dot{\theta}^2 + mgl\left(1 - \frac{\theta^2}{2}\right).$$

1.2 (2)

The Euler–Lagrange equation is:

$$\frac{\partial L}{\partial \theta} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right),$$

which gives:

$$-mgl\theta = ml^2\ddot{\theta}.$$

Rearranging, we get:

$$\ddot{\theta} = -\frac{g}{l}\theta.$$

With the initial conditions $\theta(0) = \theta_0$ and $\dot{\theta}(0) = 0$, we have:

$$\theta(t) = \theta_0 \cos\left(\sqrt{\frac{g}{l}}\, t\right).$$

From this, we know that the period is:

$$T_\theta = 2\pi\sqrt{\frac{l}{g}}.$$

2 II

2.1 (3)

$$I = \int_0^{2a} \lambda \xi^2 \, d\xi = \frac{m}{2a} \cdot \frac{8a^3}{3} = \frac{4}{3} m a^2.$$

2.2 (4)

The kinetic energy is:

$$K = \frac{1}{2}\left(\frac{4}{3}ma^2\right)\dot{\theta}^2 = \frac{2}{3}ma^2\dot{\theta}^2,$$

and the potential energy is:

$$V = -mga\cos\theta \approx -mga\left(1 - \frac{\theta^2}{2}\right).$$

Thus, the Lagrangian is:

$$L = K - V \approx \frac{2}{3}ma^2\dot{\theta}^2 + mga\left(1 - \frac{\theta^2}{2}\right).$$

2.3 (5)

Applying the Euler–Lagrange equation as before:

$$-mga\theta = \frac{4}{3}ma^2\ddot{\theta},$$

which gives:

$$\ddot{\theta} = -\frac{3g}{4a}\theta.$$

Thus, the period is:

$$T_\theta = 2\pi\sqrt{\frac{4a}{3g}}.$$

Comparing this with the simple pendulum case, we find that the equivalent length is:

$$l_{\text{eq}} = \frac{4}{3}a.$$

3 III

The spring constant is $k = \frac{mg}{6l}$.

At the static equilibrium, we have:

$$\frac{1}{6}mg = k(a - l_0),$$

where l_0 is the natural length of the spring. Substituting the given value of k, we get:

$$l_0 = a - l.$$

3.1 (6)

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 2a\sin\theta + (a+r)\sin\phi \\ -2a\cos\theta - (a+r)\cos\phi \end{pmatrix}.$$

Taking the time derivatives:

$$\begin{pmatrix} \dot{X} \\ \dot{Y} \end{pmatrix} = \begin{pmatrix} 2a\dot{\theta}\cos\theta + (a+r)\dot{\phi}\cos\phi + \dot{r}\sin\phi \\ 2a\dot{\theta}\sin\theta + (a+r)\dot{\phi}\sin\phi - \dot{r}\cos\phi \end{pmatrix}.$$

3.2 (7)

$$V = V_g + V_s,$$

where V_g is the gravitational potential energy, and V_s is the elastic potential energy of the spring.

Dropping terms of order higher than two, we finally get:

$$V = mg\left(\frac{2}{3}a\theta^2 + \frac{1}{12}a\phi^2 + \frac{1}{12l}r^2\right) + mg\left(\frac{l}{12} - \frac{3a}{2}\right).$$

3.3 (8)

As for the kinetic energy, we write:

$$T = \frac{1}{2}I\dot{\theta}^2 + \frac{m}{12}(\dot{X}^2 + \dot{Y}^2).$$

Since we only need the squares of $\dot{X}$ and $\dot{Y}$ up to second order, and neither velocity has a zeroth-order term, it is sufficient to keep only their first-order terms before squaring.

More generally, consider a position function $R(q_1, \ldots, q_n)$ with no explicit time dependence. By the chain rule, we have:

$$\dot{R} = \sum_i \frac{\partial R}{\partial q_i}\dot{q}_i.$$

Thus, $\dot{R} = 0$ when all $\dot{q}_i = 0$. This means that, when expanding around $(q_i, \dot{q}_i) = (0, 0)$, there is no zeroth-order term in $\dot{R}$.

In our case, keeping only the first-order terms:

$$\dot{X} \approx 2a\dot{\theta} + a\dot{\phi}, \qquad \dot{Y} \approx -\dot{r}.$$

Finally, we get:

$$T = \frac{2}{3}ma^2\dot{\theta}^2 + \frac{m}{12}\left(4a^2\dot{\theta}^2 + a^2\dot{\phi}^2 + 4a^2\dot{\theta}\dot{\phi} + \dot{r}^2\right).$$

3.4 (9)

Now we can write the Lagrangian:

$$\begin{aligned} L = T - V = {} & \frac{2}{3}ma^2\dot{\theta}^2 \\ & + \frac{m}{12}\left(4a^2\dot{\theta}^2 + a^2\dot{\phi}^2 + 4a^2\dot{\theta}\dot{\phi} + \dot{r}^2\right) \\ & - mg\left(\frac{2}{3}a\theta^2 + \frac{1}{12}a\phi^2 + \frac{1}{12l}r^2\right) + C. \end{aligned}$$

Here, C is just a constant that we can neglect when applying the Euler–Lagrange equation.

For r, we have:

$$\ddot{r} = -\frac{g}{l}r,$$

which means that the period of r is:

$$T_r = 2\pi\sqrt{\frac{l}{g}}.$$

3.5 (10)

Finally, applying the Euler–Lagrange equations to θ and ϕ, we get:

$$\begin{cases} -\dfrac{4g}{a}\theta = 6\ddot{\theta} + \ddot{\phi}, \\ -\dfrac{g}{a}\phi = 2\ddot{\theta} + \ddot{\phi}. \end{cases}$$

Given $Q_1 = 4\theta + \phi$ and $Q_2 = 2\theta - \phi$, we can combine the two equations to obtain:

$$\ddot{Q}_1 = -\frac{g}{2a}Q_1, \qquad \ddot{Q}_2 = -\frac{2g}{a}Q_2.$$

Thus, we know that the equivalent lengths are:

$$l_{Q_1} = 2a, \qquad l_{Q_2} = \frac{a}{2}.$$